

UMUT TULAY

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EDUCATION

University of Minnesota Twin Cities

Graduation Date: May 2025

Bachelor of Arts in Computer Science

GPA: 3.9

Relevant Courseworks: Algorithms and Data Structures, Artificial Intelligence, Operating Systems, Machine Learning Fundamentals, Computer Security, Computer Networks, Machine Arch., Program Design and Development, Adv. Programming Princip., Probability

Accomplishments/Projects: High Distinction, Dean's List (All eight semesters), Tau Sigma Honor Society, Autograder Project in C, Visual Transit Simulator Project in Java, Wordle Game Project in Python, Lisp Interpreter Project in OCaml

SKILLS

Languages: Java, Python, C, Typescript, JavaScript, CSS, C++, OCaml, Assembly, AHK, SQL (Postgres), Go, Turkish, English, French

Frameworks: Playwright, Pytest, Node.js, Express.js, MongoDB, React, JUnit, Spring Boot, PyTorch

Developer Tools: Github, Git, Linux, IntelliJ, VS Code, PyCharm, Maven, Gradle, Vim, LaTeX, Shell

Certifications: Akuna Capital Options 101, Akuna Capital Options 201

EXPERIENCE

Research Assistant (Reference: Nathan Torunsky, PhD Candidate, torun005@umn.edu)

February 2025 – Present

University of Minnesota Twin Cities

Minneapolis, MN

- Build a website in JavaScript to unify 3 behavioral experiments for the Decision Making Lab, improving experiment accessibility

Teaching Assistant (Reference: Bernardo Bianco Prado, Teaching Specialist, 7153388156)

September 2024 – May 2025

University of Minnesota Twin Cities

Minneapolis, MN

- Guided 5+ students weekly on C programming concepts, coding projects and linear algebra homework, addressing challenges in debugging and algorithm design skills which enhanced project quality and comprehension
- Led weekly discussion sections for 30+ students, offering live coding demonstrations and interactive support, leading to a 20% increase in lab completion and higher student performance
- Assessed 12 homework, 4 projects, and 3 exams per semester, providing targeted feedback, improving performance by 10% on average

Software Engineer

June 2024 – August 2024

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Ankara, Turkey

- Configured a relational database (PostgreSQL) on Raspberry Pi, optimizing data storage efficiency by 30% for IoT devices
- Optimized data transfer protocols in Python, lowering operating costs by 30%
- Collaborated with a 4-person dev team to develop and deploy client-specific IoT database solutions, achieving 100% on-time delivery

PROJECTS

Food Waste Reduction Website (Hackathon 5th Place) | [GitHub Link](#) | *Rust, TypeScript, CSS, React*

- Developed a full-stack web application to minimize food waste by connecting restaurants to nearby food shelters and individuals
- Deployed a Rust backend, integrating Google Maps API to fetch real-time restaurant and shelter locations
- Designed a responsive React front-end engineering with TypeScript and CSS for a seamless user experience

Mental Math Practice Website | [Website Link](#) | *JavaScript, CSS, Express.js, Node.js, Vercel*

- Created a website using JavaScript, Node.js, and Express to practice arithmetic operations with time-constrained questions
- Programmed an Interactive UI with CSS to provide real-time feedback on correct answers and tracking performance

Multithreaded Image Matching Server | [GitHub Link](#) | *C*

- Launched and implemented a multithreaded client-server system in C using POSIX threads
- Engineered a server capable of receiving user-submitted images, processing requests concurrently, and matching images

RANKINGS

Placed 5th in Hackathon coding competition, collaborating in a team to develop a full stack web application addressing food waste

Placed 5th in Association for Computing Machinery (ACM) UMN's Capture The Flag competition

Placed 1st at Century College and 2nd in Minnesota in the AMATYC Math League

Ranked in the 99.6th percentile in the YKS (Turkish university entrance exam)